FIELD GUIDE / AGRICULTURE INTELLIGENCE

FieldPulse
User Guide

A beginner-friendly guide to the irrigation lab, AI-style helpers, farm calculators, field notebook, reports and learning workspace.

01
MEASURE
Sensor readings

02
VALIDATE
Check the inputs

03
DECIDE
Apply safety rules

04
ACT + LOG
Record the result

In one sentence: FieldPulse is a digital notebook and planning toolkit. It helps you work with information you enter; it does not see your farm or control farm equipment.

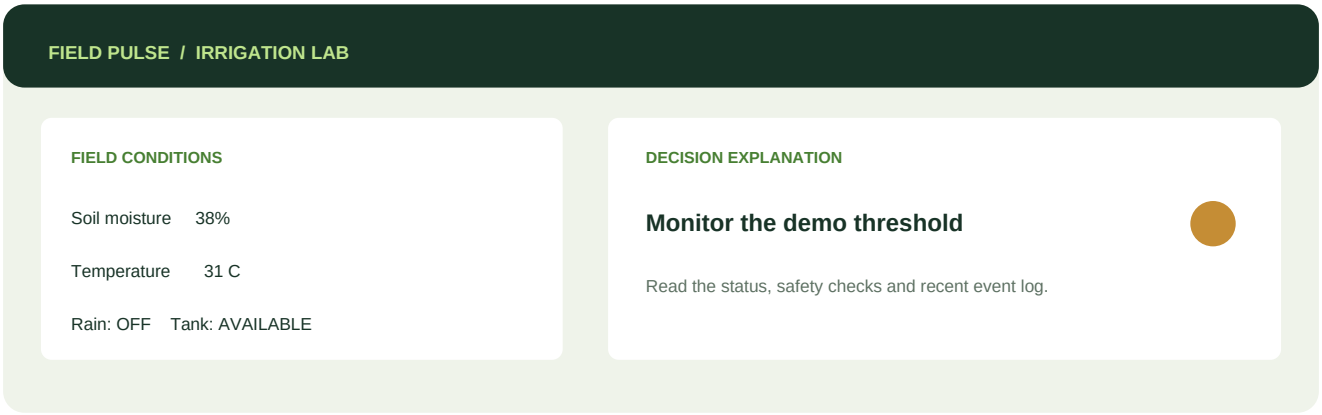
Project: FieldPulse Smart Irrigation Concept **Team:** Phoenix Coders

Prepared for first-time users. Read the limitations and safety notes before using outputs to plan real farm operations.

01 / GETTING STARTED

Find your way around FieldPulse

The website is arranged as separate workspaces. Choose a section from the navigation bar at the top. Only the selected workspace is shown, so you can focus on one task at a time.



The eight workspaces

WORKSPACE	WHAT IT DOES	WHEN TO OPEN IT
Overview	Starting page with links to the rest of the site.	When you open FieldPulse or want a shortcut.
Irrigation Lab	Runs the soil-moisture and safety-rule simulator.	To learn how the example irrigation rules respond.
AI Studio	Provides eight local helpers and content generators.	To organise field observations and next checks.
Farm Tools	Holds 15 practical calculators.	To estimate water, energy, seed, nutrients and other quantities.
Field Records	Saves typed observations in this browser.	To keep a local field notebook.
Reports	Compares entered baseline and prototype measurements.	To create a TXT, CSV or print-ready report.
Learn	Explains the system, features and safety limitations.	Before using calculations or planning hardware.
Team	Shows project identity and contributor links.	To learn about the project team.

Helpful controls

• **Search:** Select the magnifying-glass button or press the slash key (/). Search by a word such as seed, sensor, crop scout or pump. Select a result to open its workspace.

• **Light / dark mode:** Select the sun or moon button. The selected colour theme is saved in this browser when storage is available.

• **Guided tour:** Select the play button or press G. The tour opens each workspace, highlights an area and explains what to notice and how to use it.

• **Keyboard shortcuts:** / opens search, D changes theme, G starts the tour and Esc closes a dialog or exits the tour.

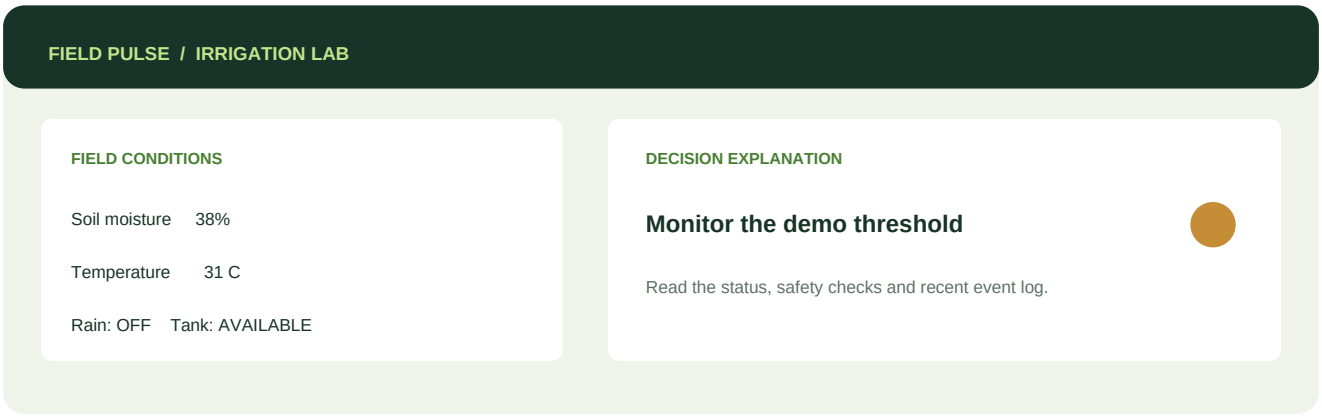
• **Mobile menu:** On a small screen, open the menu button to see the workspace navigation.

Privacy note: Field notebook entries are stored in the browser on the current device. They are not synced to a server. Export CSV or JSON backups if you need to keep them.

02 / IRRIGATION LAB

Try the irrigation simulator

The simulator demonstrates a simple decision path: read conditions, validate sensor input, apply safety interlocks, compare a moisture index with an example threshold, then show a request and event log.



What each control means

CONTROL	PLAIN-ENGLISH EXPLANATION
Soil moisture	Move the slider to change the example 0 to 100 moisture index. It is not guaranteed to be a physical percentage of water in the soil.
Air temperature	Adds environmental context. The demo does not use temperature alone to request irrigation.
Crop demo profile	Chooses an illustrative threshold for the simulator. These values are placeholders, not crop prescriptions.
Control mode	Automatic applies the example rules. Manual mode lets you make a simulated ON or OFF request.
Rain detected	When enabled, the example policy pauses irrigation.
Water available	When disabled, a safety interlock blocks the pump request to represent dry-run prevention.
Sensor reading valid	When disabled, the simulator blocks a request until the input is checked.
Recalculate decision	Runs the rules again and adds an event to the log.
Event log	Shows recent demo decisions. Use Export CSV to save a copy, or Clear to remove the visible entries.

A safe way to experiment

- Set the crop demo profile and leave the simulator in Automatic mode.
- Lower the moisture slider and select Recalculate decision. Read the decision title and explanation.
- Turn on Rain detected and recalculate. Notice that the example rule holds the request.
- Turn off rain and mark Water available as unavailable. The safety interlock should block the request.
- Switch to Manual mode and try a request. The safety interlocks still apply in this demonstration.

Real-world warning: This page never operates a physical pump. A real controller needs electrical review, correctly rated isolated switching, dry-run and overcurrent protection, maximum runtime, safe default-off behaviour and supervised tests. A browser page must not be the essential safety controller.

03 / AI STUDIO

Eight helpers for field thinking

The AI Studio uses local, deterministic rules, pattern checklists and phrase templates. It does not call a trained AI model, fetch live field data or provide expert diagnosis. It is designed to help you organise information and decide what to inspect next.

HELPER	WHAT IT DOES	IMPORTANT LIMIT
Farm Copilot	Uses crop, stage, moisture index, temperature, rain status, water availability and your notes to produce a prioritised action brief.	A checklist prompt, not an irrigation command or agronomic prescription.
Crop Symptom Scout	Organises the symptoms you tick, estimates the share of plants affected and suggests evidence to collect.	It does not identify a disease or recommend pesticide treatment.
Sensor & Pump Detective	Compares three entered probe values and expected versus measured flow, then flags large differences.	A difference may have many causes. It cannot confirm a leak or failed sensor.
Crop Rotation Planner	Creates a crop-family diversity worksheet and prompts you to check soil, season, water and market constraints.	It does not choose a crop suitable for a specific district or farm.
Farm Task Generator	Builds a checklist for irrigation, scouting, soil records, harvest or general field rounds.	Edit every task to suit your crop, equipment and local advice.
Voice-Ready Helper	Creates sample English, Hindi or Punjabi phrases for a future voice interface.	It does not listen, record audio, translate arbitrary speech or control equipment.
Field Resilience Playbook	Builds a preparation, during-event and recovery checklist for heat, heavy rain, drought, power loss or limited water.	It is not a forecast or emergency warning. Follow official local alerts and safety guidance.
Harvest & Post-harvest Planner	Creates a harvest handover checklist for maturity checks, labour, transport, drying, segregation and storage.	It does not determine harvest maturity or safe moisture and storage standards.

How to use a helper

- Open AI Studio from the top navigation.
- Choose the tab that matches the task you want to organise.
- Enter only what you have actually observed. Do not invent readings or tick symptoms that are not present.
- Select the generate or inspect button and read the explanation, not just the label.
- Compare the result with field checks, local extension guidance and verified measurements before taking action.
- Save useful observations in Field Records so you can compare them later.

Why the site calls these AI-style helpers: They make a response from the information you provide, but they do not learn from farm data or use an external AI model. A production AI service would need model integration, secure data handling, evaluation, clear limitations and human review.

04 / FARM TOOLS

Calculators for water, soil and crops

Every calculator works from numbers entered by the user. Read the description and formula before using its output in a farm plan.

TOOL	WHAT IT CALCULATES	INPUTS / HOW TO READ IT
01. Water Requirement	Converts area and a net water depth into estimated gross volume using application efficiency.	Area (ha), depth (mm), efficiency and events per week. 1 mm over 1 hectare = 10,000 L.
02. Pump Runtime & Energy	Estimates runtime, electricity use and indicative energy cost for a target water volume.	Volume (L), flow (L/min), input power (W) and tariff (INR/kWh). Runtime = volume / flow.
03. Sensor Calibration	Maps a current analogue reading between your dry and wet reference values to a 0-100 index.	Dry ADC, wet ADC and current ADC. A normalized index is not laboratory volumetric water content.
04. Multi-Zone Planner	Estimates the time and volume for identical zones that run one after another.	Zone count, minutes per zone, flow per active zone and cycles per week. Assumes a constant flow.
05. Root-Zone Interval	Estimates a root-zone water buffer and divides it by user-entered daily water use.	Field capacity, wilting point, root depth, allowed depletion and daily use. All soil inputs need local validation.
06. Rainwater Harvest	Estimates gross collection from a rainfall depth and a catchment surface.	Rainfall (mm), area (m2), runoff coefficient and storage capacity. Actual usable collection is lower.

Unit reminders

- **mm:** millimetre, a depth of water. One millimetre across one square metre is one litre.
- **ha:** hectare. One hectare is 10,000 square metres.
- **L/min:** litres per minute. It is a flow rate, not a stored volume.
- **kWh:** kilowatt-hour, a unit of electrical energy. A pump power rating alone does not determine the real farm energy bill.

Important: The root-zone interval is a simplified planning estimate, not a watering schedule. Soil texture, rooting depth, crop stage, weather, irrigation method and sensor placement all matter.

05 / FARM TOOLS

Calculators for inputs, equipment and costs

The remaining tools convert an input plan or measurement into a quantity. They do not create the agronomic recommendation behind that input.

TOOL	WHAT IT CALCULATES	CHECK BEFORE USING
07. Seed Quantity	Adjusts a target seed rate for seed-lot germination and purity, then scales it to field area.	Use a locally suitable rate and a current seed-lot test. Verify units and variety.
08. Fertilizer Product	Converts a target nutrient quantity into product mass using the nutrient grade on the bag.	The tool does not recommend nutrient rates. Use a soil test, product label and local advice.
09. Hydraulic Power	Estimates idealised pump power from flow, total head and entered efficiency.	Actual selection requires pump curves, pipe losses, motor efficiency and an engineering review.
10. Label-Based Spray Mix	Multiplies a rate copied from a product label by tank volume and tank count.	Never use the calculator to choose a product or rate. Follow the exact label, PPE, local law and safety intervals.
11. Scouting Incidence	Calculates the percentage of inspected plants that show a selected symptom.	Incidence is a monitoring metric, not a pest threshold or treatment trigger.
12. Investment Payback	Estimates simple payback from an initial cost and entered annual net benefit.	A simple scenario only. It omits discounting, financing, price changes, risk and unexpected repairs.

Useful formulas in plain English

CALCULATOR	FORMULA USED
Seed quantity	Target seed rate x area / (germination fraction x purity fraction).
Fertilizer product	Target nutrient kg / (product grade percent / 100).
Spray label arithmetic	Label quantity per litre x tank volume x number of tanks.
Scouting incidence	Affected plants / inspected plants x 100.
Simple payback	Initial investment / (annual benefit - annual running cost), only if the net benefit is positive.

Spray safety: The spray calculator is only arithmetic for a rate already printed on the exact product label. It does not tell you whether a product is allowed, suitable, safe or needed. Follow local regulations and ask a qualified adviser when in doubt.

06 / ADDITIONAL FARM TOOLS

Three extra tools for field planning

These tools were added to help users think beyond a basic irrigation timer: daily crop-water balance, water-distribution checks and harvest storage capacity. Their inputs still require local validation.

TOOL	WHAT IT CALCULATES	HOW TO USE THE RESULT
13. Daily Crop Water Balance	Subtracts effective rainfall and a soil-water contribution from crop water use entered by the user. Converts the net deficit to gross water depth and a field volume.	Use a trusted crop-water estimate and locally measured inputs. It is a simplified daily balance, not a full crop evapotranspiration model.
14. Irrigation Uniformity Checker	Compares four catch-container measurements and calculates a lower-quarter distribution uniformity percentage.	Use equal containers and a consistent test duration. A four-point demo is only a screening calculation; a proper field test needs more sample points.
15. Harvest Storage Space	Estimates storage volume from harvest mass, bulk density and a planned fill fraction.	Use a measured or reliable crop-specific bulk density and check the actual storage design, ventilation and safety requirements.

A few more plain-English formula notes

Daily water balance: Net deficit = the larger of zero or crop water use minus effective rain minus the soil contribution. Gross depth then divides by the entered application efficiency.

Uniformity: For the four sample values, lower-quarter distribution uniformity = the smallest catch divided by the average catch, multiplied by 100. Repeat the test method consistently and use an accepted protocol for a real evaluation.

Storage volume: Required volume = harvest mass in kilograms divided by bulk density, then divided by the planned fill fraction. Bulk density changes with crop, moisture, variety and handling.

Use these tools to ask better questions, not to skip field checks. A result is only as useful as the measurements and assumptions entered.

07 / RECORDS AND REPORTS

Keep evidence, then explain it

Field Records and Reports are designed to help you keep track of what happened and communicate the numbers you entered.

Field Notebook

The Field Records workspace lets you save a date, field name, record type and free-text observation. Examples include crop observations, soil tests, irrigation events, scouting, weather notes, input costs, equipment checks and harvest notes.

• Enter a date, field or zone name, choose the record type and describe what you observed or did.

• Select Save field entry. The item appears in the record list.

• Use the search box and type filter to find a record later.

• Export CSV for use in a spreadsheet or JSON for a structured backup.

• Delete an item only when you are sure you no longer need it.

Reports Centre

The report builder compares baseline and prototype water and energy totals, along with the trial duration, manual interventions and notes. It generates a summary of the differences and a validation checklist.

ACTION	RESULT
Generate report	Displays water and energy changes, trial length, notes and a validation checklist.
Download TXT	Saves a readable text report that can be opened in a text editor.
Download CSV	Saves comparison rows that can be opened in spreadsheet software.
Print / Save PDF	Opens the browser print dialog. Choose a printer or “Save as PDF” when available.

How to interpret the comparison: If prototype water is lower than baseline in the entered figures, the report describes that difference. It does not prove the prototype caused the difference or that the result will repeat. Compare similar crop stages, field areas, weather and irrigation methods, using measured volumes and energy values.

Learn and Team

The Learn workspace has the system flow, feature explanations, staged build plan and common safety questions. The Team workspace identifies Phoenix Coders and the project contributors.

08 / RESPONSIBLE USE

Know what the website cannot do

FieldPulse is an educational front-end prototype. It is not a certified controller, agronomist, diagnostic laboratory, financial adviser or live farm-data platform. Use it to learn, organise and estimate - then verify with appropriate people and measurements.

Current limits

• **No live sensors:** The site does not connect to an ESP32, Arduino, moisture probe, flow meter or weather station.

• **No pump control:** The pump state is text in a simulator. It does not switch a relay or valve.

• **No live weather:** Rain and temperature are values you enter or move in the demo.

• **No trained AI:** The AI Studio is a local collection of rules, checklists and phrase templates.

• **No cloud database:** Field notebook entries stay in the current browser. Export backups.

• **No guaranteed outcomes:** Calculator results and report comparisons do not prove water savings, yield gains, pest control or financial return.

Before a physical prototype

• Choose and calibrate sensors for the actual soil, salinity, installation depth and field conditions.

• Validate every sensor and safety interlock independently. Test safe behaviour during a reset, power loss, invalid sensor input and empty tank.

• Use a pump-rated, correctly isolated driver. Never connect a pump motor directly to a microcontroller pin.

• Add dry-run protection, overcurrent protection, maximum run-time cut-off, secure wiring and a safe default-off state.

• Use waterproof enclosures and an electrical design review appropriate to the equipment and local requirements.

• Begin with a supervised, low-risk bench test. Move to a measured field pilot only after safe bench tests pass.

• Record baseline and prototype water use, energy use, crop stage, rainfall, soil moisture, pump runtime and manual interventions under comparable conditions.

Remember: The strongest next step is not to add more automation. It is to validate the sensors, prove the safety interlocks, record real measurements and evaluate the system carefully.

Project identity

FieldPulse - Smart Irrigation & Agriculture Planning Concept
Team: Phoenix Coders
Team leader: Aryan Das
Team member: Asmi Jha

End of guide. Use the Learn workspace in the website for quick explanations, and the guided tour to visit each workspace in order.